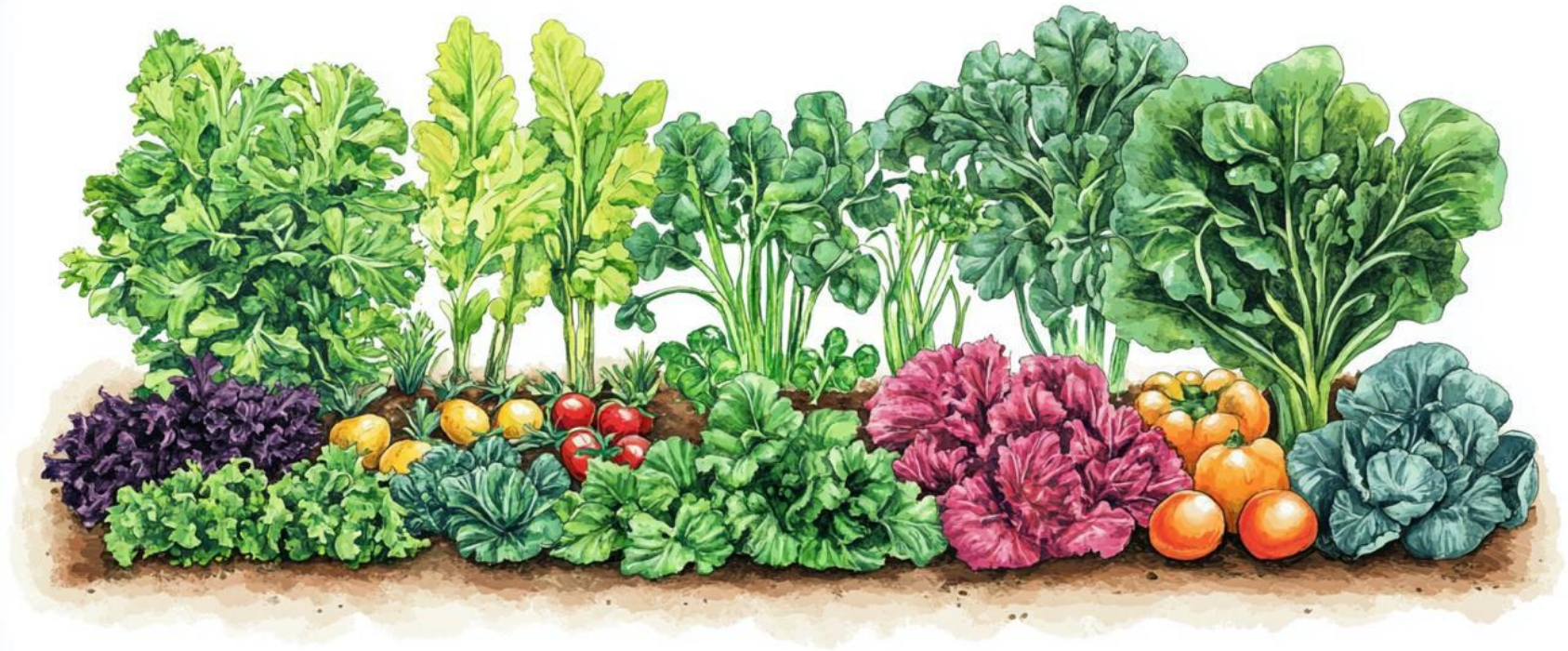
NumPy



NumPy – huh, yeah – what's it good for?

- NumPy introduces a new data structure: **the array**

> An array is a regular, N-dimensional grid of data of the same type, typically numerical data

- Great for storing homogeneous data, where every element in the array has the same meaning. E.g. images, sound, time series

Efficient machine-native implementation

- Data is stored in a contiguous chunk of memory, using machine-native data types
- Separate metadata tells numpy how to interpret that memory as an array

Memory block storage

0	1	2	3	4	5	6	7	8

int64
8 bytes

NumPy array metadata

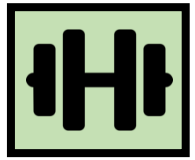
dtype	int64
ndim	2
shape	(3, 3)
strides	(24, 8)

NumPy view

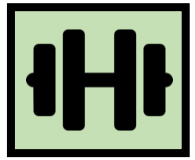
0	**1**	**2**
3	**4**	**5**
6	**7**	**8**

int64
8 bytes

Is NumPy any better than a list-of-lists?



Is NumPy any better than a list-of-lists?



- Numpy arrays are implemented using machine-native data types, which means that common operations and algorithms can be implemented with less per-element overhead, making them faster than a Python list-of-lists

- Faster in what sense?
- **It's not going to scale any better than list-of-lists**
  Scaling complexity is the same. E.g., square matrix multiplication is still $O(N^3)$
- **Much faster for fixed-size problems**
- This speed advantage strictly depends on operations being made in C and Fortran. Avoid Python for-loops and use existing NumPy and SciPy functionality!

[*] The best matrix multiplication algorithm as of 2025 scales as O(n**2.371339). However, out-of-the-box NumPy packages usually use a Fortran library called OpenBLAS that implements a O(n**3) algorithm

How does NumPy implement array operations?

x

NumPy array metadata

dtype	int64
ndim	2
shape	(3, 3)
strides	(24, 8)
data	

Memory block storage

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**

NumPy view

0	**1**	**2**
3	**4**	**5**
6	**7**	**8**

int64
8 bytes

```
y = x.mean(axis=1)
```

For many operations, NumPy creates a **new memory block** with the result

x

NumPy array metadata

dtype	int64
ndim	2
shape	(3, 3)
strides	(24, 8)
data	

Memory block storage

0	1	2	3	4	5	6	7	8

Array layout

0	1	2
3	4	5
6	7	8

int64
8 bytes

y = x.mean(axis=1)

NumPy array metadata

dtype	float64
ndim	1
shape	(3,)
strides	(8,)
data	

Memory block storage

1.0	4.0	7.0

Array layout

1.0	4.0	7.0

float64
8 bytes

How could this operation be implemented?

```
x = np.arange(9)
```

NumPy array metadata

dtype	int64
ndim	1
shape	(9,)
strides	(8,)
data	

Memory block storage

0	1	2	3	4	5	6	7	8

Array layout

0	1	2	3	4	5	6	7	8

int64
8 bytes

```
y = x.reshape((3, 3))
```

1) With a **copy** of the original data

```
x = np.arange(9)
```

NumPy array metadata

dtype	int64
ndim	1
shape	(9,)
strides	(8,)
data	

Memory block storage

0	1	2	3	4	5	6	7	8

Array layout

0	1	2	3	4	5	6	7	8

int64
8 bytes

```
y = x.reshape((3, 3))
```

NumPy array metadata

dtype	int64
ndim	2
shape	(3, 3)
strides	(24, 8)
data	

Memory block storage

0	1	2	3	4	5	6	7	8

Array layout

0	1	2
3	4	5
6	7	8

int64
8 bytes

1) With a **copy** of the original data

```
x = np.arange(9)
```

NumPy array metadata

dtype	int64
ndim	1
shape	(9,)
strides	(8,)
data	

Memor

0	1	2	3	4	5	6	7	8

How does **copying** scale if the original array has N elements?

yout

2	3	4	5	6	7	8

```
y = x.reshape((3, 3))
```

NumPy array metadata

dtype	int64
ndim	2
shape	(3, 3)
strides	(24, 8)
data	

Memory block storage

0	1	2	3	4	5	6	7	8

Array layout

0	1	2
3	4	5
6	7	8

int64
8 bytes

1) With a **copy** of the original data

```
x = np.arange(9)
```

NumPy array metadata

dtype	int64
ndim	1
shape	(9,)
strides	(8,)
data	

Memor

0	1

How does **copying** scale if the original array has N elements? O(N)

Array layout

2	3	4	5	6	7	8

```
y = x.reshape((3, 3))
```

NumPy array metadata

dtype	int64
ndim	2
shape	(3, 3)
strides	(24, 8)
data	

Memory block storage

0	1	2	3	4	5	6	7	8

Array layout

0	1	2
3	4	5
6	7	8

int64
8 bytes

2) With a **view** of the original data

```
x = np.arange(9)
```

NumPy array metadata

dtype	int64
ndim	1
shape	(9,)
strides	(8,)
data	

Memory block storage

0	1	2	3	4	5	6	7	8

Array layout

0	1	2	3	4	5	6	7	8

int64
8 bytes

```
y = x.reshape((3, 3))
```

NumPy array metadata

dtype	int64
ndim	2
shape	(3,3)
strides	(24, 8)
data	

Array layout

0	1	2
3	4	5
6	7	8

int64
8 bytes

2) With a **view** of the original data

```
x = np.arange(9)
```

NumPy array metadata

dtype	int64
ndim	1
shape	(9,)
strides	(8,)
data	

Memor

0	1

Array layout

2	3	4	5	6	7	8

How does **creating a view** scale if the original array has N elements?

```
y = x.reshape((3, 3))
```

NumPy array metadata

dtype	int64
ndim	2
shape	(3,3)
strides	(24, 8)
data	

Array layout

0	1	2
3	4	5
6	7	8

int64
8 bytes

2) With a **view** of the original data

```
x = np.arange(9)
```

NumPy array metadata

dtype	int64
ndim	1
shape	(9,)
strides	(8,)
data	

Memor

0	1

Array layout

0	1	2	3	4	5	6	7	8

How does **creating a view** scale if the original array has N elements?
O(1)!

```
y = x.reshape((3, 3))
```

NumPy array metadata

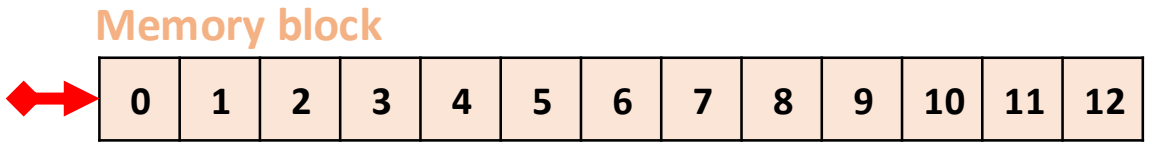
dtype	int64
ndim	2
shape	(3,3)
strides	(24, 8)
data	

Array layout

0	1	2
3	4	5
6	7	8

int64
8 bytes

O(1) operations in NumPy



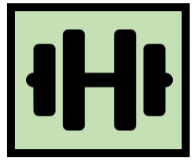
NumPy operation	NumPy array metadata	Array layout
`x`	ndim: 2 shape: (4, 3) strides: (24, 8) data: ◆	0 1 2 3 4 5 6 7 8 9 10 11
`y = x.ravel()`	ndim: 1 shape: (12,) strides: (8,) data: ◆	0 1 2 3 4 5 6 7 8 9 10 11
`y = x.T`	ndim: 2 shape: (3, 4) strides: (8, 24) data: ◆	0 3 6 9 1 4 7 10 2 5 8 11
`y = x.reshape((2, 6))`	ndim: 2 shape: (2, 6) strides: (48, 8) data: ◆	0 1 2 3 4 5 6 7 8 9 10 11

The golden rule of NumPy

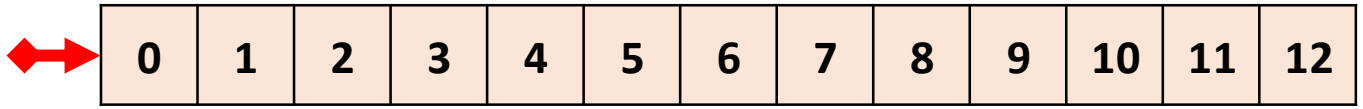
Operations that can be executed by only changing the metadata return a “**view “ of the original data memory block**

In all other cases, it creates a **“copy”** with a new data memory block

Views *vs.* copies in indexing operations



Memory block



NumPy operation

```
x
```

NumPy array metadata

dtype	int64
ndim	2
shape	(4, 3)
strides	(24, 8)
data	◆

0	**1**	**2**
3	**4**	**5**
6	**7**	**8**
9	**10**	**11**

Slicing

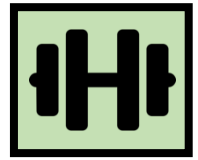
```
x[::3, ::2]
```

dtype	
ndim	
shape	
strides	
data	◆

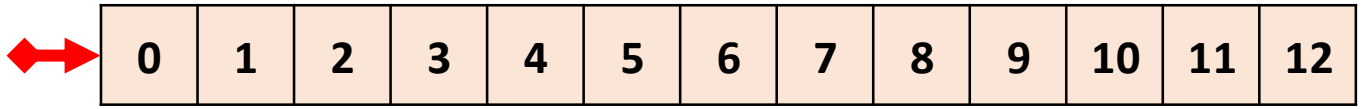
0	**2**
9	**11**

Can this operation be executed by changing the metadata?

Views *vs.* copies in indexing operations



Memory block



NumPy operation

`x`

NumPy array metadata

dtype	int64
ndim	2
shape	(4, 3)
strides	(24, 8)
data	◆

0	**1**	**2**
3	**4**	**5**
6	**7**	**8**
9	**10**	**11**

Slicing

`x[::3, ::2]`

dtype	int64
ndim	2
shape	(2, 2)
strides	(72, 16)
data	◆

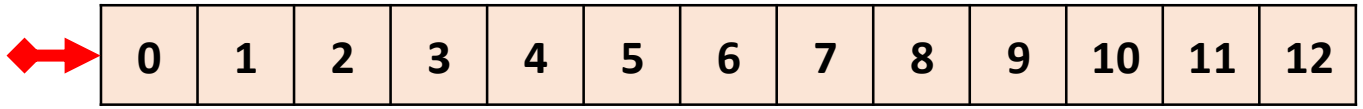
0	**2**
9	**11**

Can this operation be executed by changing the metadata? YES!

Can it be done for all slicing operations?

Views *vs.* copies in indexing operations

Memory block



NumPy operation

x

NumPy array metadata

Slicing always returns a **view** of the original data

Slicing

```
x[::3, ::2]
```

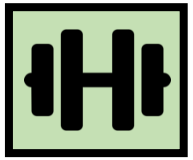
dtype	int64
ndim	2
shape	(2, 2)
strides	(72, 16)
data	◆

0	**2**
9	**11**

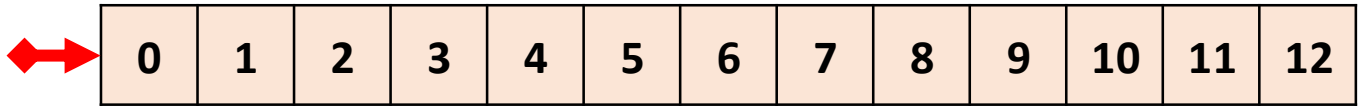
Can this operation be executed by changing the metadata?
YES!

Can it be done for all slicing operations?
YES!

Views *vs.* copies in indexing operations



Memory block



NumPy operation

x

NumPy array metadata

dtype	int64
ndim	2
shape	(4, 3)
strides	(24, 8)
data	◆

0	1	2
3	4	5
6	7	8
9	10	11

Fancy indexing

```
x[[0, 3, 2], [1, 1, 2]]
```

row indices

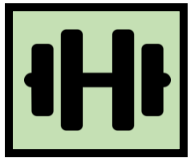
column indices

dtype	
ndim	
shape	
strides	
data	

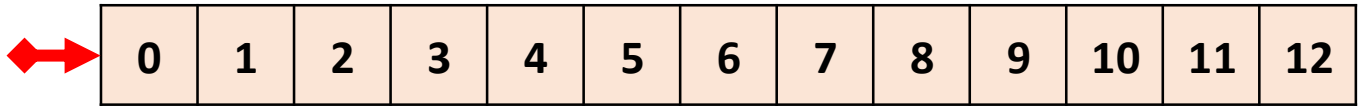
1	10	8

Can this operation be executed by changing the metadata?

Views *vs.* copies in indexing operations



Memory block



NumPy operation

x

NumPy array metadata

dtype	int64
ndim	2
shape	(4, 3)
strides	(24, 8)
data	◆

0	**1**	**2**
3	**4**	**5**
6	**7**	**8**
9	**10**	**11**

Fancy indexing

```
x[[0, 3, 2], [1, 1, 2]]
```

row indices

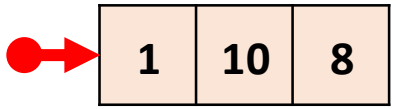
column indices

dtype	int64
ndim	1
shape	(3,)
strides	(8,)
data	●

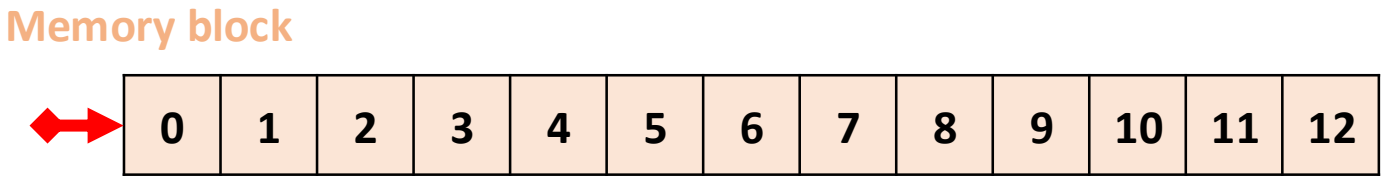
1	**10**	**8**

Can this operation be executed by changing the metadata?
NO!

Memory block



Views *vs.* copies in indexing operations



NumPy operation

x

NumPy array metadata

Fancy indexing

`x[[0, 3, 2],`

row indices

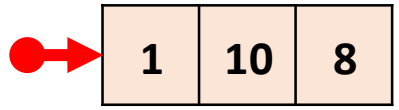
column indices

strides	(8,)
data	●

Fancy indexing always returns an array pointing to a **new memory block** (we still call this a "copy")

ation be executed
g the metadata?
NO!

Memory block



View or copy? Quiz

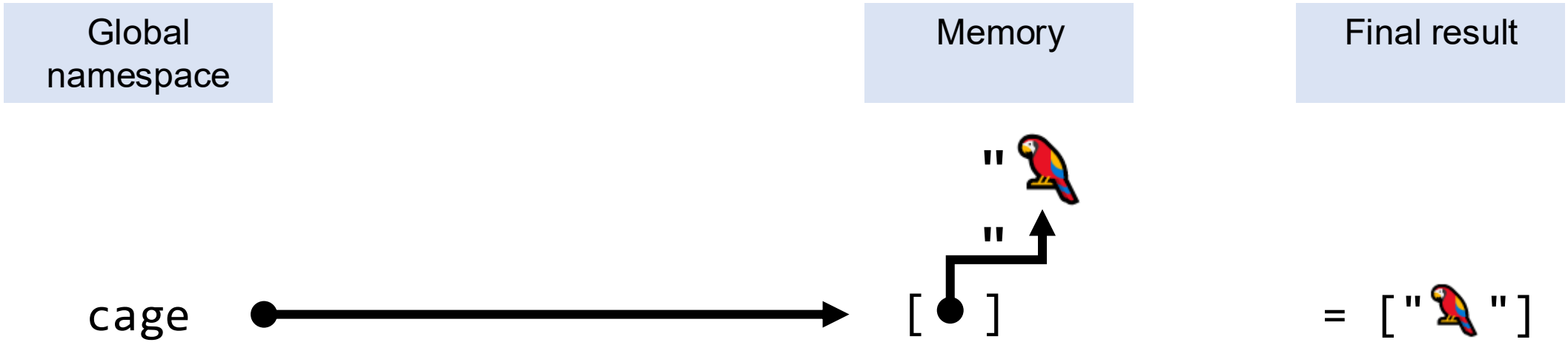
Exercise

Setting up a pet shop

Live Coding
notebooks/020_numpy/
015_assignment_by_reference.ipynb

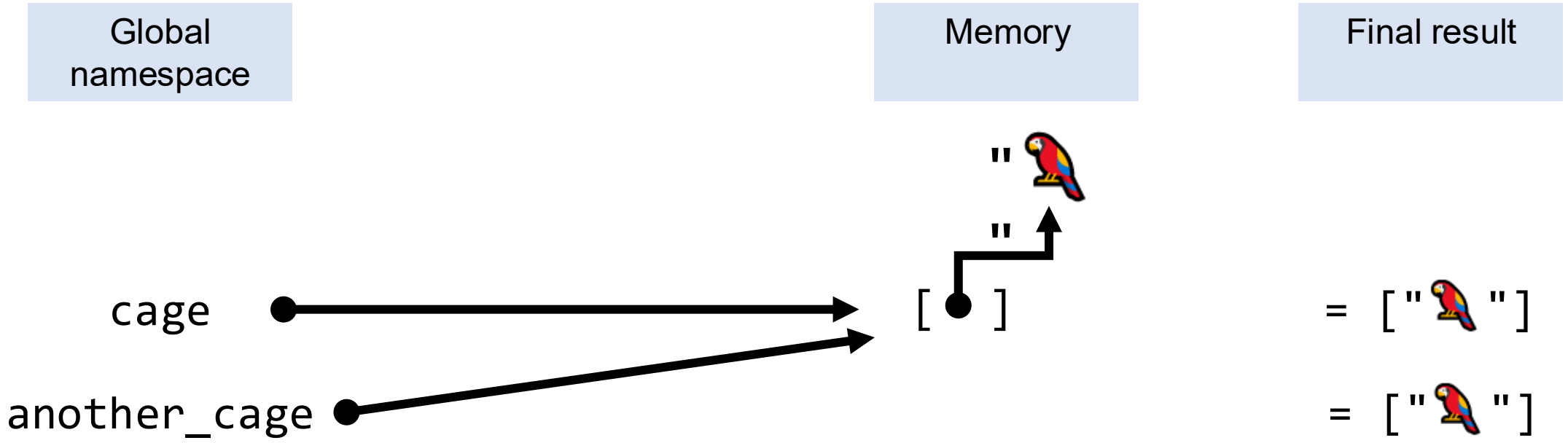
Python assignment is “by reference”

```
cage = ["🦜"]
```



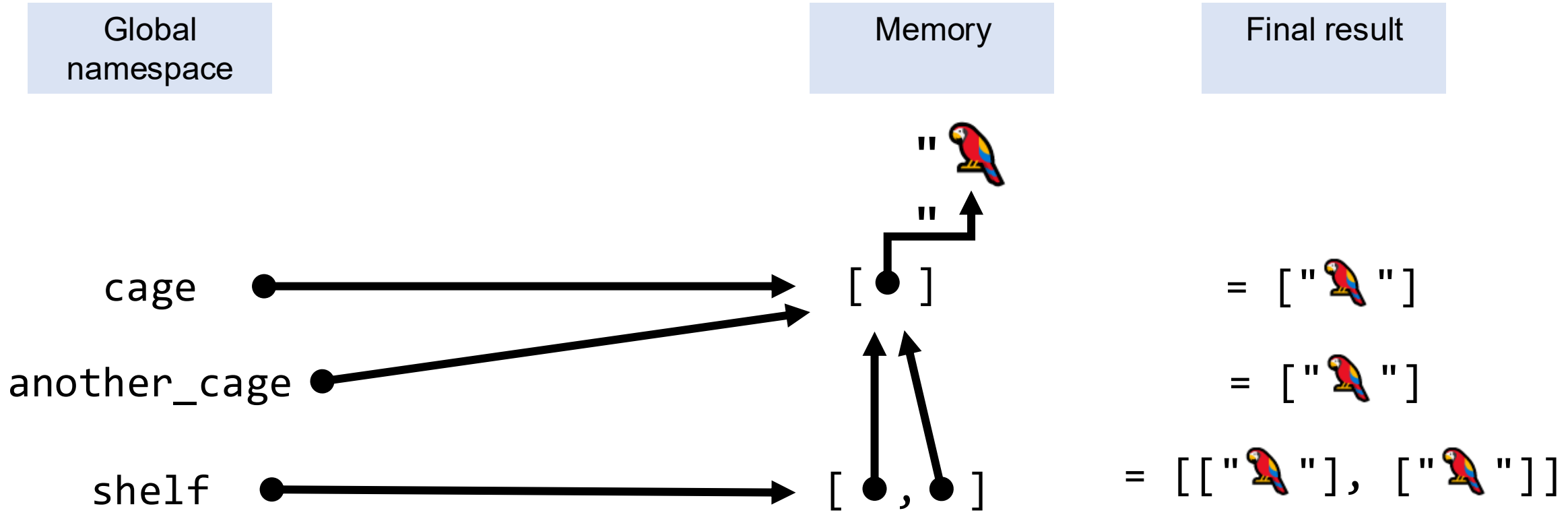
Python assignment is “by reference”

```
cage = ["🦜"]
another_cage = cage
```



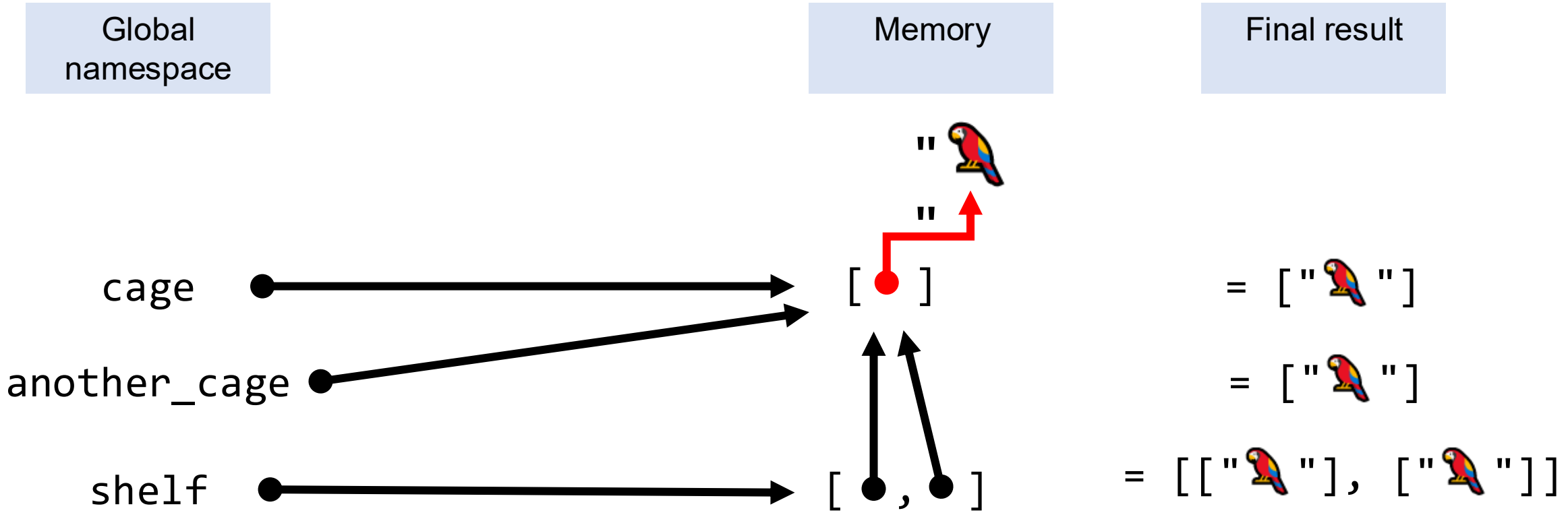
Python assignment is “by reference”

```
cage = ["🦜"]
another_cage = cage
shelf = [cage, cage]
```



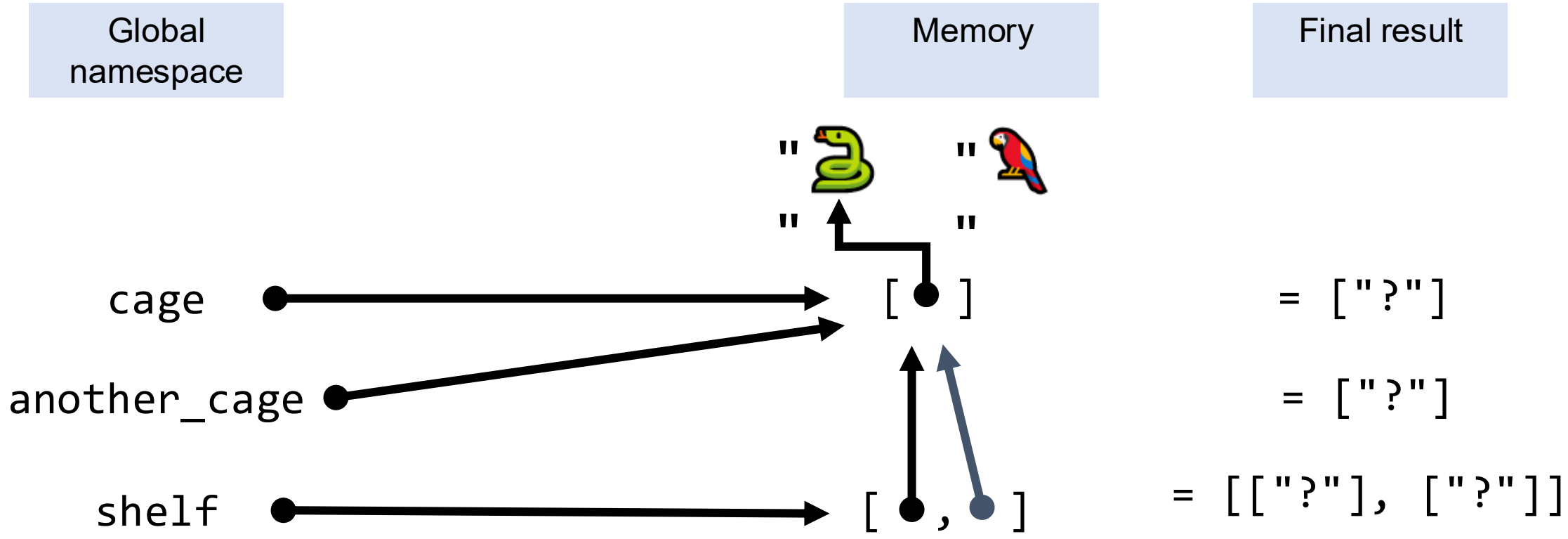
Python assignment is “by reference”

```
cage = ["🦜"]
another_cage = cage
shelf = [cage, cage]
cage[0]
```



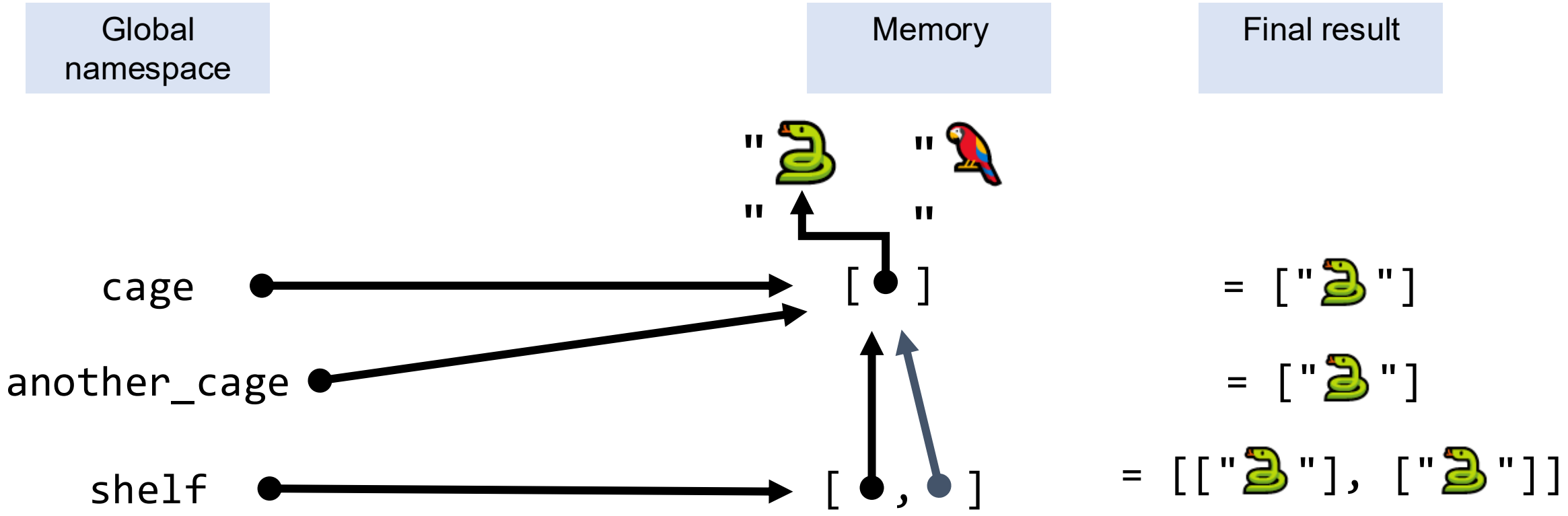
Python assignment is “by reference”

```
cage = ["🦜"]
another_cage = cage
shelf = [cage, cage]
cage[0] = "🐍"
```



Python assignment is “by reference”

```
cage = ["🦜"]
another_cage = cage
shelf = [cage, cage]
cage[0] = "🐍"
```



Changing one view changes them all

```
x = np.arange(12).reshape(4,3)
```

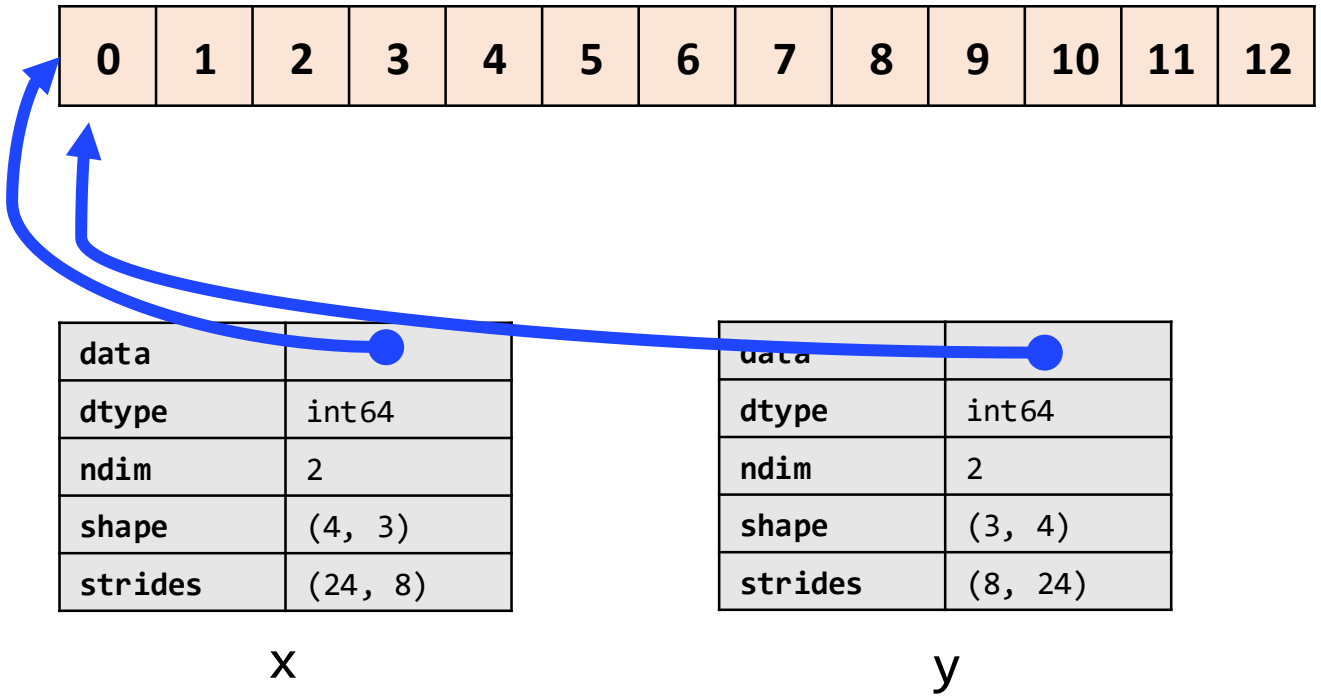
```
x
```

```
array([[ 0,  1,  2],
       [ 3,  4,  5],
       [ 6,  7,  8],
       [ 9, 10, 11]])
```

```
y = x.T
```

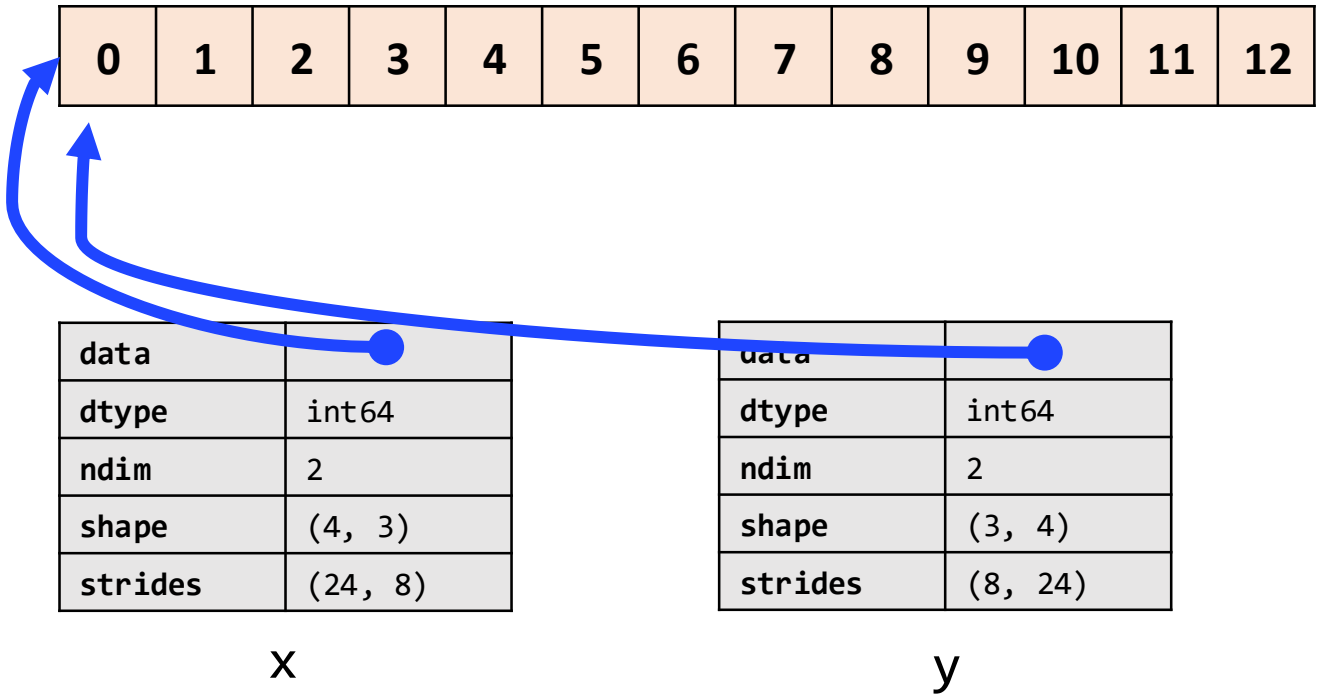
```
y
```

```
array([[ 0,  3,  6,  9],
       [ 1,  4,  7, 10],
       [ 2,  5,  8, 11]])
```



Changing one view changes them all

```
y[0, 0] = 13
```



Changing one view changes them all

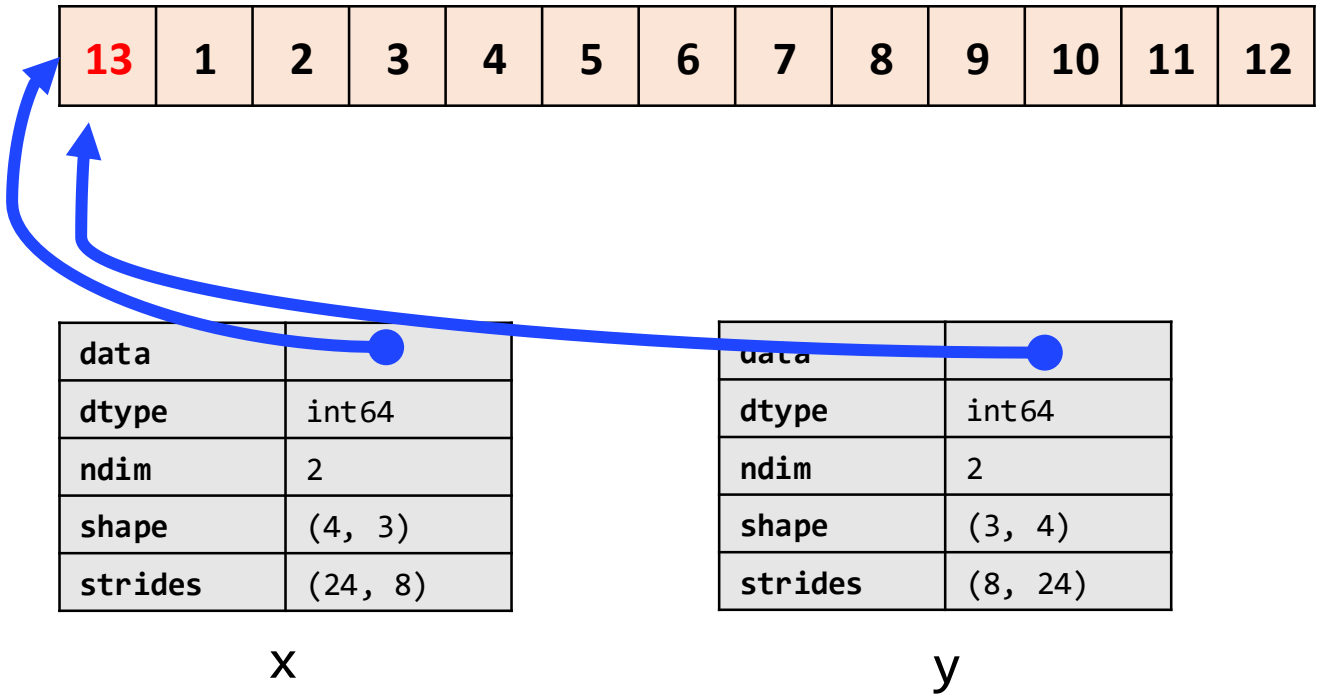
```
y[0, 0] = 13
```

```
y
```

```
array([[13,  3,  6,  9],
       [ 1,  4,  7, 10],
       [ 2,  5,  8, 11]])
```

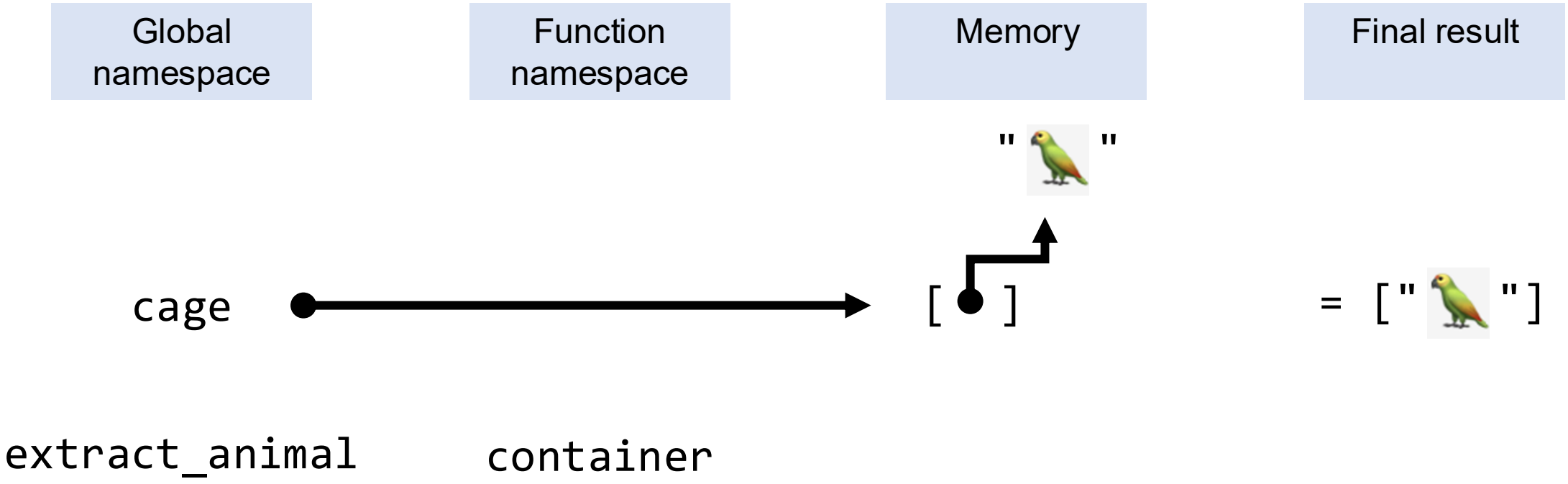
```
x
```

```
array([[13,  1,  2],
       [ 3,  4,  5],
       [ 6,  7,  8],
       [ 9, 10, 11]])
```



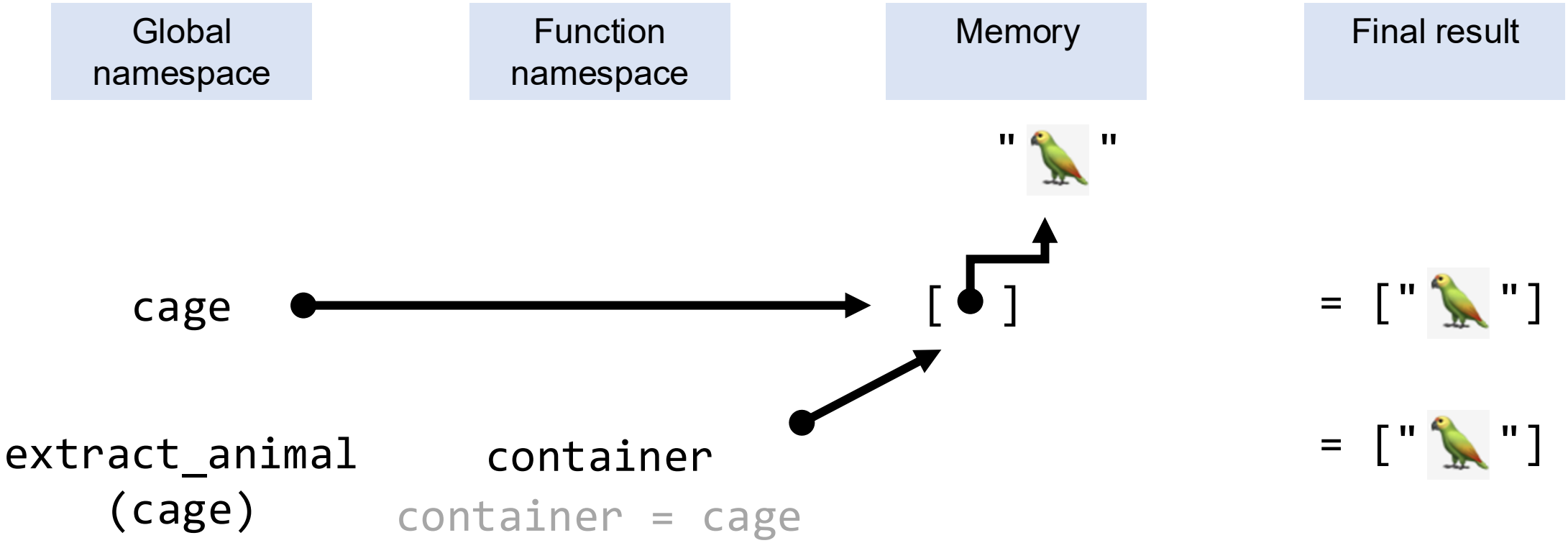
Passing an argument is an assignment

```
cage = ["🦜"]
def extract_animal(container):
        return container.pop()
```



Passing an argument is an assignment

```
cage = ["🦜"]
def extract_animal(container):
        return container.pop()
extract_animal(cage)
```



Passing an argument is an assignment

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cage = ["🦜"]
def extract_animal(container):
        return container.pop()
extract_animal(cage)
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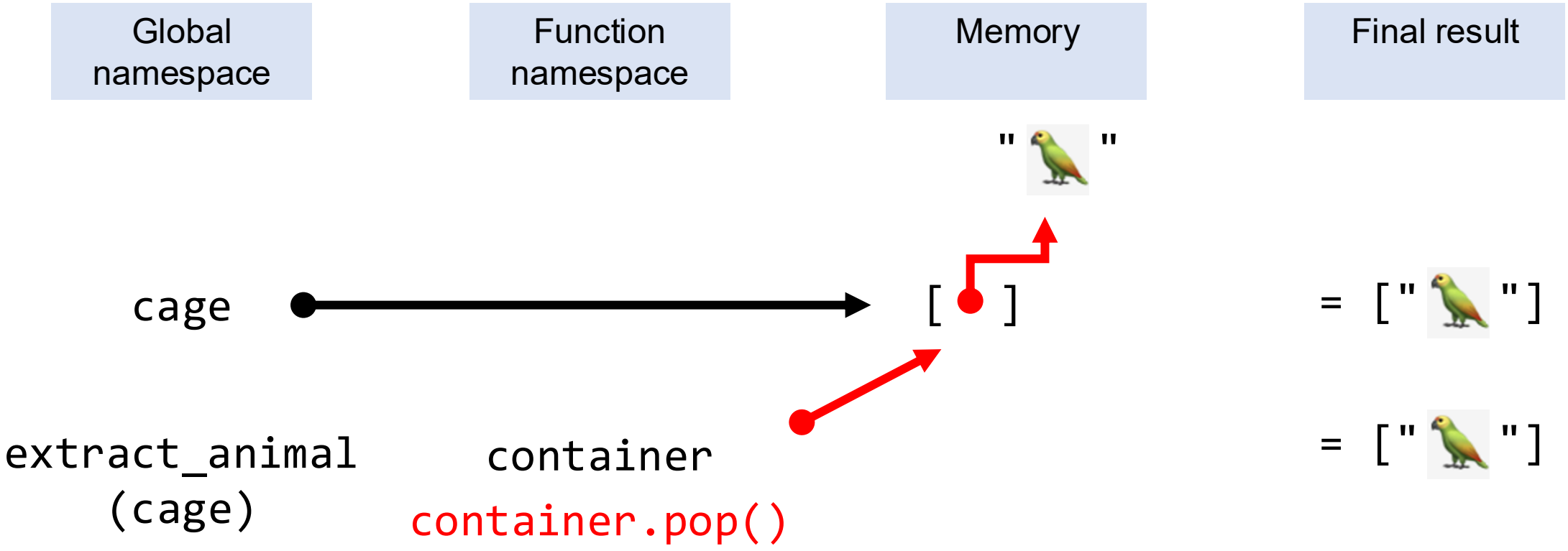

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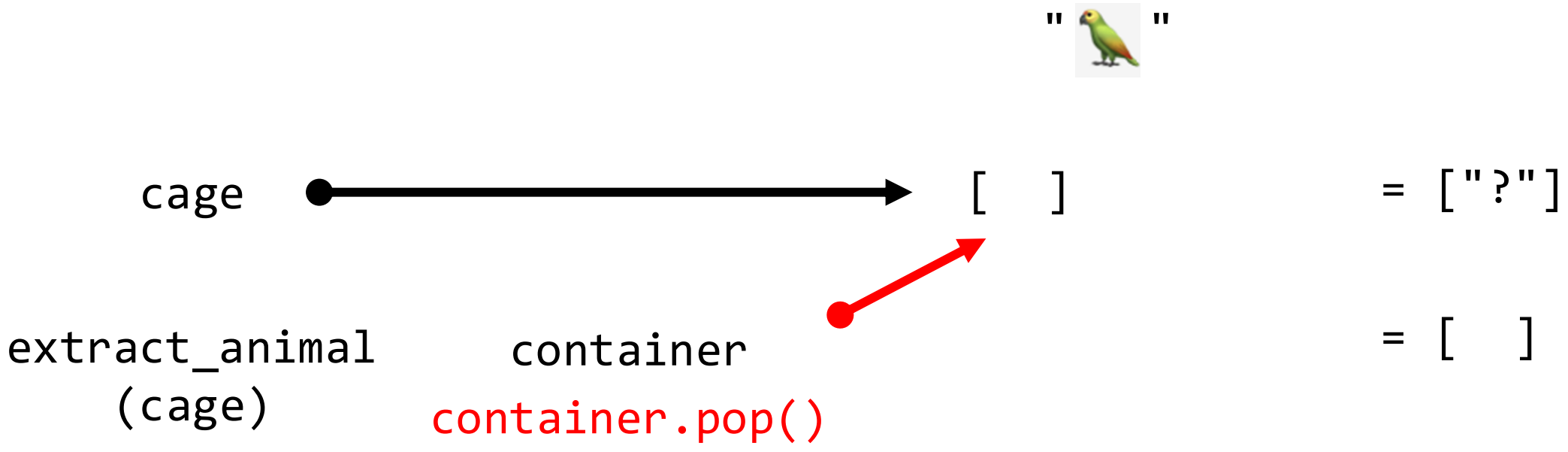
Global namespace	Function namespace	Memory	Final result



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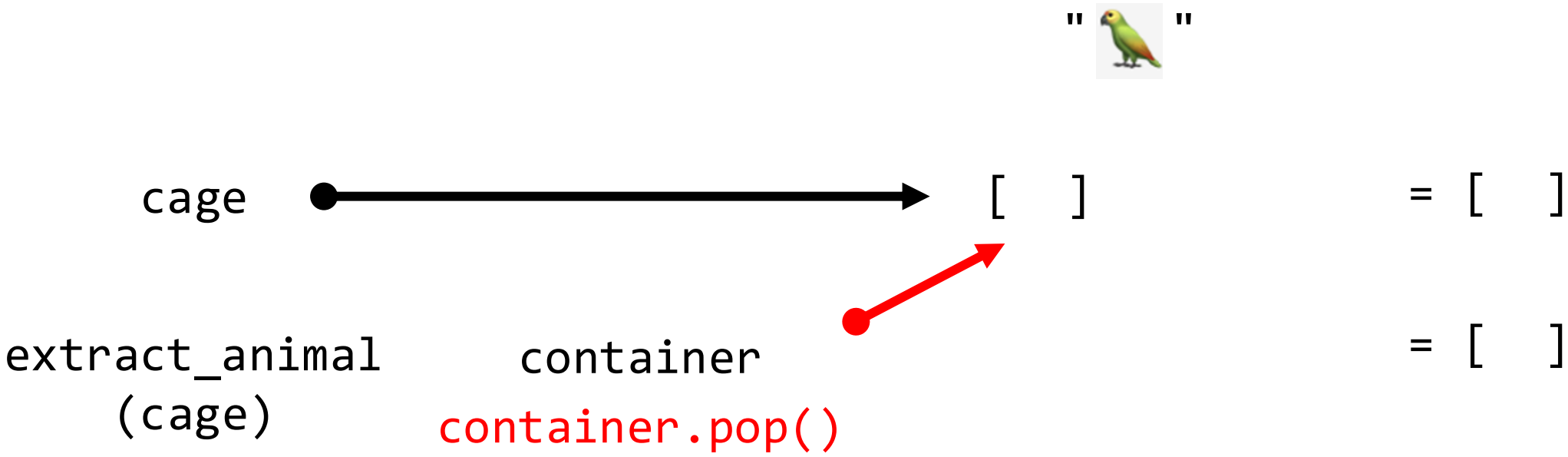
Global namespace	Function namespace	Memory	Final result



Assignments and functions

Exercise

`exercises/numpy_assignments/side_effects.ipynb`

Put up the GREEN sticker after you solve part A

Careful with your functions

```python
def robust_log(x, cte=1e-10):
    """ Compute the log of the elements of an array.

    Values that are equal to 0.0 in `x` are substituted with a tiny constant `cte`
    to avoid a divide-by-zero warning, and `-inf` values in the output arrays.
    """
    x[x == 0] = cte
    return np.log(x)
```

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    """
    x[x == 0] = cte
    return np.log(x)
```

```python
a = np.array([[0.3, 0.01], [0, 1]])
print(a)
```

```
[[0.3  0.01]
 [0.   1.  ]]
```

```python
# Using the NumPy's log directly
np.log(a)
```

```
/tmp/ipykernel_50294/3282750587.py:2: RuntimeWarning: divide by zero encountered in log
  np.log(a)
```

```
array([[-1.2039728 , -4.60517019],
       [      -inf,  0.        ]])
```

```python
# Our function handles values equal zero to return a small value
robust_log(a)
```

```
array([[ -1.2039728 ,  -4.60517019],
       [-23.02585093,   0.        ]])
```

Careful with your functions

```
def robust_log(x, cte=1e-10):
    """ Compute the log of the elements of an array.

    Values that are equal to 0.0 in `x` are substituted with a tiny constant `cte`
    to avoid a divide-by-zero warning, and `-inf` values in the output arrays.
    """
    x[x == 0] = cte
    return np.log(x)
```

```
a = np.array([[0.3, 0.01], [0, 1]])
b = a[1, :]  # A view of `a`
print(b)
```

```
[0. 1.]
```

```
robust_log(b)
```

```
array([-23.02585093,   0.        ])
```

```
b
```

```
array([1.e-10, 1.e+00])
```

```
a
```

```
array([[3.e-01, 1.e-02],
       [1.e-10, 1.e+00]])
```

The input array has been modified!

… and so have all other views of the same data!

Careful with your functions

Best practice: functions that take an array as an input should avoid modifying it in place!
Always make a copy or be super extra clear in the docstring

```python
def robust_log(x, cte=1e-10):
    """ Compute the log of the elements of an array.

    Values that are equal to 0.0 in `x` are substituted with a tiny constant `cte`
    to avoid a divide-by-zero warning, and `-inf` values in the output arrays.
    """
    x = x.copy()
    x[x == 0] = cte
    return np.log(x)
```

NumPy views and copies summary

View

- There can be multiple views of the same memory block, interpreted as different arrays
- Slicing returns a view
- In-place operations on a view modify the memory block and all of its views

Copy (incl. new data)

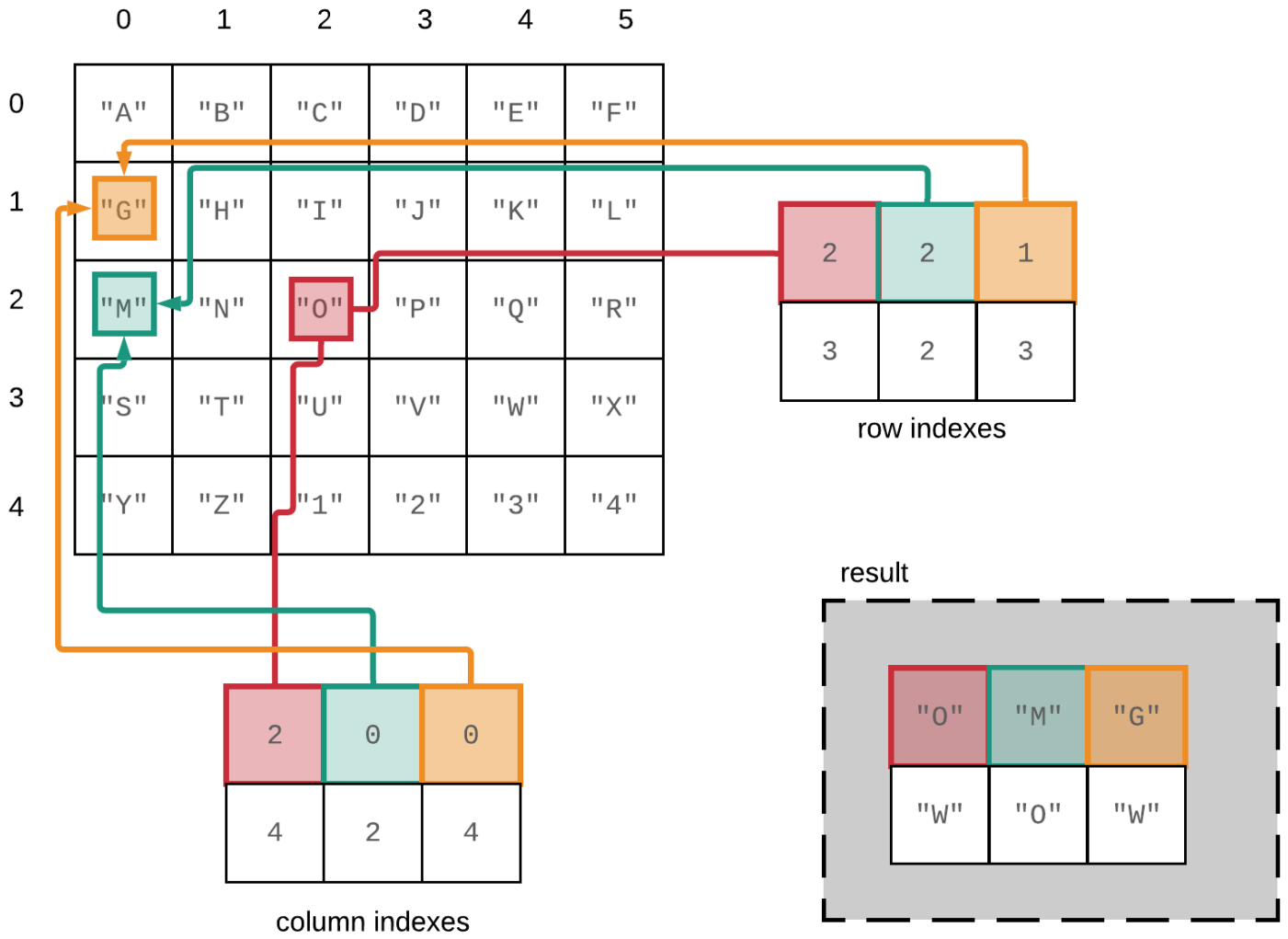
- When a copy of an array needs to be created, it allocates a separate memory block and associates it with new metadata
- Fancy indexing always returns copies
- A copy can be forced with .copy()

Up next: Tabular Data

Data, v4.0

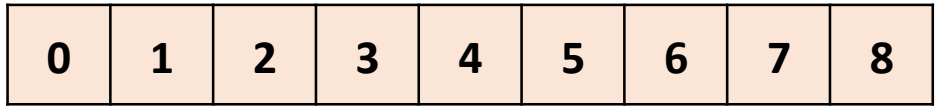
Fancy indexing in NumPy – reference slide

A[[2, 2, 1], [2, 0, 0]]



A special kind of view: broadcasting operations

Memory block



NumPy array metadata

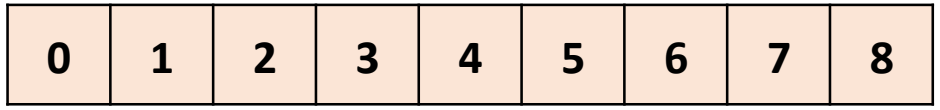
dtype	int64
ndim	2
shape	(4, 9)
strides	(0, 8)

The shape says we have 4 rows and 9 columns

A stride of 0 means that for each new row, we don't move in memory

A special kind of view: broadcasting operations

Memory block



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dtype	int64
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shape	(4, 9)
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The shape says we have 4 rows and 9 columns

A stride of 0 means that for each new row, we don't move in memory

As a result, we obtain a view with duplicated rows, without using extra memory!

NumPy view

0	1	2	3	4	5	6	7	8
0	1	2	3	4	5	6	7	8
0	1	2	3	4	5	6	7	8
0	1	2	3	4	5	6	7	8

NumPy uses broadcasting to perform operation on arrays of different shape without having to allocate extra memory